

COMPONENTS

High-power chip resistors

Ideal for compact power supply and motion control applications, the WHP-C0508X, WHP-C0612X, and WHP-C1020X chip resistors feature inverse geometry with the terminations on the long sides. This gives the new WHP chip resistors very high thermal contact with the PCB and low thermal impedance. As a result, it is possible to support two to three times conventional power ratings for the same footprint without relying on excessive PCB copper heatsink areas. For example, the WHP-C1020 with terminal heatsinking copper areas of 80mm² runs at 2W continuous dissipation.

TT Electronics

www.ttelectronics.com

Battery management ICs

The new family of multi-cell full battery front-end (BFE) ICs for battery management systems (BMS) is built for the larger, high-voltage battery packs that power e-scooters, energy storage, high-voltage power tools and other high-voltage equipment. The new ICs provide fast and flexible cell balancing up to 200mA +, a critical function to enable fast recharging and high utilisation in large battery packs with hot plug tolerance up to 62V. They also feature extensive built-in self-diagnostics and high integration for simplifying the design cycle and reducing customers' system BOM costs significantly, accelerating design and BOM selection time from months to weeks.

Renesas Electronics

www.renesas.com



Pressure sensor IC

The new compact high-accuracy barometric pressure sensor IC, BM1390GLV (-Z), featuring a waterproof rating of IPX8, is ideal for home appliances, industrial equipment, and compact IoT devices. The BM1390GLV combines MEMS and control circuitry with original waterproof technology in a compact 2.0mm × 2.0mm × 1.0mm package. A proprietary built-in temperature compensation function ensures immunity against temperature fluctuations while its ceramic package minimises fluctuations of device-specific characteristics caused by mechanical stress during board mounting. These features enable high-accuracy barometric pressure measurement in applications requiring waterproof performance.

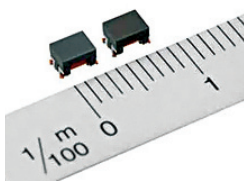
ROHM Semiconductor

www.rohm.com



Compact automotive inductors

The newly developed ADL3225VM inductors for use in automotive power over coax (PoC) systems provide a compact solution for designers looking to reduce vehicle weight as manufacturers add more sensors and cameras to accommodate expanding automotive and advanced driver-assistance system (ADAS) applications. The proprietary structural design and wire-winding manufacturing process ensure high impedance over a broad bandpass of 1MHz up to 1GHz. The inductors are compliant with AEC-Q200 and achieve the highest rated current in the industry



in a 3.2mm × 2.5mm × 2.5mm (L × W × H) size.

TDK Corporation

www.tdk.com

USB-C controller

The EZ-PD BCR (barrel connector replacement) is a highly integrated USB-C controller. Together with the USB-C connector, it replaces barrel connectors, custom connectors, or legacy USB connectors in electronic devices.

The EZ-PD BCR solution supports the USB power delivery (PD) standard that interoperates with all USB-C power adaptors without the need of firmware development. Changing to a USB-C charging port using EZ-PD BCR at the power sink offers higher levels of convenience to consumers by being agnostic to connector types. Manufacturers can benefit from the consolidation of large variants of chargers to a standard USB-C charger.

Infineon Technologies

www.infineon.com



Long range sensor

The AR2700 high-speed long range sensor is an ultra-compact rangefinder that is eye-safe and has a measurement



frequency of up to 40kHz. It can measure natural targets up to 70 metres away or retroreflective targets up to 270 metres away. The

AR2700 measures distance with a rapidly-pulsing collimated laser beam that creates an infrared spot on a target surface. With an ultracompact design, the AR2700 is used by equipment manufacturers for applications that demand very high sampling rates such as for detecting objects in industrial automation, or for